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Late Cretaceous geomagnetic polarity sequence: detailed rock and palaeomagnetic studies of the Scaglia Rossa limestone at Gubbio, Italy

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Summary. A magnetic polarity sequence has been developed for the Late Cretaceous from about 500 stratigraphic samples of the Scaglia Rossa limestone from a 250-m thick section near Gubbio, Italy. Although a large soft component is present in all samples, it is possible to remove it by alternating field magnetic cleaning. A modification of the minimum dispersion technique was used to determine the representative direction for each sample. The mean cleaned directions of samples with negative and positive polarities are slightly different indicating incomplete removal of a stable remanence component. Isothermal remnanence acquisition curves indicate that magnetite is present in all samples, and haematite also in most of them. The NRM directions of samples containing magnetite only are not significantly different from those containing both magnetite and haematite. It is proposed that the haematization was either contemporaneous with the primary magnetization or occurred very soon afterwards. The magnetizations respond positively to a fold test and therefore pre-date the Late Tertiary folding. The sedimentation in the section was continuous, although the rate in the Maastrichtian was almost double the rate earlier in the Late Cretaceous. The palaeontologically dated Gubbio section has been proposed as a type section for the Late Cretaceous (Alvarez *et al.*). Detailed sampling for the present study locates each clearly defined reversal within about 0.2 m, establishes the existence of a previously overlooked polarity zone and corrects the polarity sequence for local minor tectonic effects. The Gubbio magnetic polarity sequence shows a one-to-one correspondence with the polarity sequences derived from oceanic magnetic anomalies 29–34. A palaeontological age can therefore be assigned to each of the corresponding Late Cretaceous geomagnetic reversals.

1 Introduction

Our knowledge of the history of reversals of the geomagnetic field derives from three sources. Palaeomagnetic studies of radiometrically dated volcanic rocks (Cox, Doell & Dalrymple 1963, 1964), and of palaeontologically dated deep sea sediments (Opdyke *et al.*

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1966) established the polarity sequence for about the last 4.5 Myr. Attempts have been made to carry it back into the Miocene by overlapping cores with hiatuses at the top and correlatable biostratigraphies (Opdyke, Burckle & Todd 1974; Theyer & Hammond 1974). Recently, the polarity sequence from volcanic rocks has been extended to about 6.5 Myr in an apparently continuous series of radiometrically-dated Icelandic lavas (McDougal *et al.* 1977).

Analysis of sea-floor spreading (Vine & Matthews 1963) is the third source of geomagnetic polarity information. Vine & Wilson (1965) showed that the polarity sequence inferred from marine magnetic anomalies agreed with the observed sequence in volcanic rocks for the last 3.4 Myr and used the radiometric ages of the reversal boundaries to estimate an average rate of sea-floor spreading. The polarity sequence obtained was greatly extended by Heirtzler *et al.* (1968), who interpreted marine magnetic anomalies from the Pacific, Indian and Atlantic oceans. The young end of their sequence was dated by comparison with the radiometrically dated continental igneous sequence, and an average rate of spreading was deduced. Assuming this axial zone spreading rate to be constant for a given ocean, the distances to numbered anomalies were converted to ages of the underlying crust, and a tentative geomagnetic polarity-timescale was produced. The oldest anomalies observed (31 and 32) were dated as Late Cretaceous, and the Cretaceous–Tertiary boundary was placed between anomalies 26 and 27.

The Heirtzler polarity timescale has subsequently been refined several times (e.g. Sclater *et al.* 1974; LaBrecque, Kent & Cande 1977). The Deep Sea Drilling Project (DSDP) in several holes sampled the sediment layer as far as its contact with the underlying basalts. Radiometric dating of the basalt and palaeontological dating of the sediment immediately above basement both involve uncertainties but have been used to provide ages for certain magnetic anomalies. Reversal boundaries are then dated by linear interpolation. However, the sequence of geomagnetic polarity epochs inferred from oceanic magnetic anomalies remains largely unverified by direct measurement in continental igneous or sedimentary rocks other than for the last 6.5 Myr.

Absolute age dating by radiometric methods has certain advantages in that it allows approximate rates of spreading (and rates of sedimentation) to be determined within the limits of accuracy of the dating method. More precise relative dating is possible palaeontologically, by reason of the intricate, detailed biozonation schemes that have been developed for correlation purposes. The global nature of geomagnetic reversals has been used successfully as a correlation tool, together with palaeontological zonation for the dating of modern unconsolidated deep sea sediments (Hays & Opdyke 1967). Indurated pelagic limestones have similar advantages for magnetic stratigraphy analysis; they represent essentially continuous sedimentation and contain abundant datable fossils. Although their magnetization is extremely weak, it can be readily measured with modern computer-controlled spinner magnetometers or cryogenic magnetometers.

Previous work on the pelagic Scaglia Rossa limestone from a number of sites distributed throughout Umbria, Italy, gave consistent palaeomagnetic results (Lowrie & Alvarez 1974, 1975; Channell & Tarling 1975; VandenBerg, Klootwijk & Wonders 1977). In a pilot magnetic stratigraphy study of an essentially complete single exposure near Gubbio, ranging from Turonian to Eocene in age, 16 stratigraphically spaced sites were sampled (Alvarez & Lowrie 1974; Lowrie & Alvarez 1975). Many of these sites were reversely magnetized. The limestone possessed satisfactory magnetic properties and its sedimentology and palaeontology were known. It appeared to be a suitable section for development of a detailed magnetic stratigraphy for the Late Cretaceous based on extensive additional, closely spaced sampling.

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2 The Gubbio section of the Scaglia Rossa limestone

The depositional environment of the Umbrian Apennines is known as far back as the Upper Triassic, represented by evaporites which have been found in drillholes. A carbonate platform developed during the Early Liassic, and broke up into a basin-and-range topography by faulting in the Middle Liassic. Pelagic sedimentation ensued, lasting until the Miocene. The Scaglia Rossa limestone was deposited from the Middle Cretaceous to the Eocene, and thus contains a 50 Myr record in a fairly uniform pelagic sequence. The major structural features of Umbria are long NW–SE trending anticlines that were formed during the Apennine folding in the Late Tertiary. The region has been considered autochthonous or parautochthonous (Bortolotti *et al.* 1970), although more extensive allochthony cannot be excluded because of possible décollement on the Triassic evaporites.

The Gubbio section outcrops along the road through the Bottaccione gorge, about a kilometre from the town of Gubbio (Fig. 1). The apparently complete Scaglia Rossa limestone sequence is continuously exposed here, and was first described by Renz (1936, 1951). Its palaeontology was restudied by Luterbacher & Premoli Silva (1962, 1964) who described a new basal Palaeocene zone (*Globigerina eugubina*), at the base of which the Cretaceous–Tertiary boundary is located.

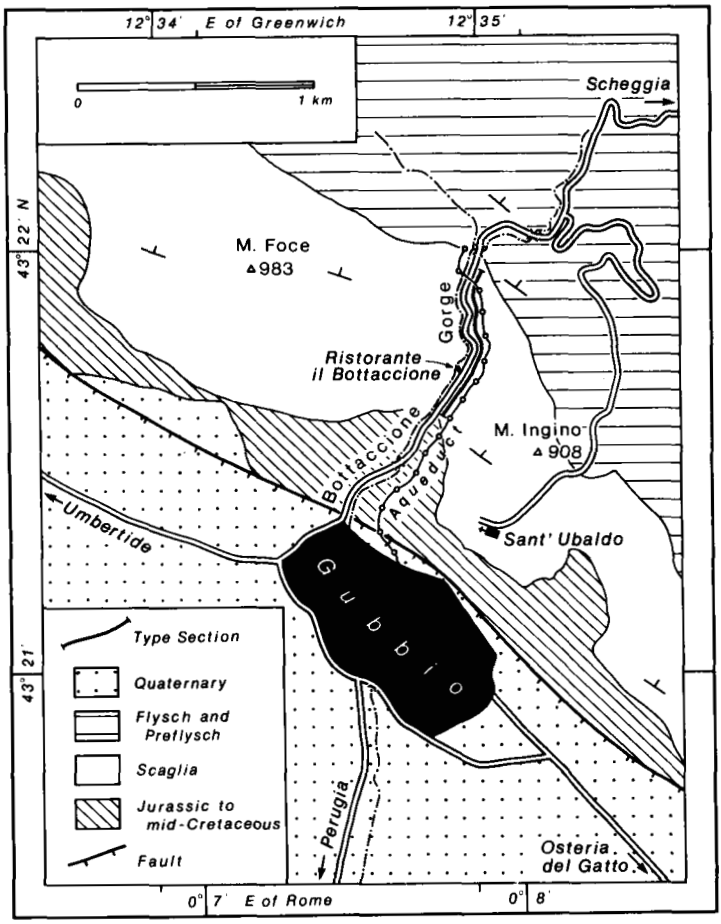


Figure 1. The Gubbio section is located along the roadcut through the Bottaccione gorge just north of the town of Gubbio in the Umbrian Apennines of Northern Italy (latitude 43° 22' N, longitude 12° 35' E).

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Table 1. Estimated sedimentation rates during the Late Cretaceous, based upon the stage boundary locations of Premoli Silva (1977) and Premoli Silva *et al.* (1977), and the timescale of van Hinte (1976).

Epoch	Boundary position (m)	Age (Myr)	Thickness (m)	Duration (Myr)	Sedimentation rate (m/Myr)
	347.6	65			
Maastrichtian			77.4	5	15.5
	270.2	70			
Campanian			74.2	8	9.3
	196	78			
Santonian			36.0	4	9.0
	~160	82			
Coniacian			24.0	4	6.0
	~136	86			
Turonian			34.5	6	5.8
	~101.5	92			

Recently, coordinated magnetic stratigraphy investigations of the Palaeocene and Upper Maastrichtian (Roggenthen & Napoleone 1977) and the Maastrichtian to Turonian (Lowrie & Alvarez 1977) in the Gubbio section have been completed and preliminary results presented. Detailed examination of the lithostratigraphy and sedimentology of the section was carried out by Arthur & Fischer (1977), whose measured section forms the framework for the magnetic stratigraphies. Palaeontological dating of the section from the Palaeocene to the top of the Santonian was carried out by Premoli Silva (1977) and extended in less detail to the base of the Turonian by Premoli Silva, Paggi & Monechi (1977). Faulting and tectonic disturbance are minor and can be taken into account in interpreting the palaeomagnetic directions. Because of the completeness of this section, its freedom from major tectonic disturbance and its importance for the Late Cretaceous and Palaeocene, it has been proposed as a magnetostratigraphic type section for this time interval (Alvarez *et al.* 1977).

The section from immediately above the Tertiary–Cretaceous boundary to the easily identifiable Bonarelli Zone (or *scisti ittiolitici*, a black organic-rich shale with fossilized fish scales) in the Turonian was sampled at approximately 1-m intervals using a portable drill with a 1-in diamond bit. On the basis of the magnetic stratigraphy obtained, further closely-spaced sampling (bed-by-bed in places) was carried out across reversal boundaries and regions of particular interest.

Using the recent timescale of van Hinte (1976) and the palaeontological zonations of the Gubbio section (Premoli Silva 1977; Premoli Silva *et al.* 1977), the sediment thicknesses of each of the major Late Cretaceous stages and appropriate sedimentation rates have been calculated (Table 1). These figures differ only slightly from the values determined by Arthur & Fischer (1977) on the basis of other timescales. The estimated average sedimentation rate for the entire Late Cretaceous section was 9.1 m/Myr. However, in the Maastrichtian the rate (15.5 m/Myr) was double that in the Turonian to Campanian stages (7.7 m/Myr). Therefore, the mean sampling interval represents about 100 000 yr, and in the intensively sampled zones about 5–10 000 yr; an individual 1-in specimen is presumed to average palaeomagnetic data for at least 1500 yr.

3 Remanent magnetization and its stability

3.1 MEASUREMENT OF REMANENCE

The remanent magnetizations were measured with a Digico complete results spinner magnetometer using a six-spin technique. Two remanent magnetization components are

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determined in each of the six spins, and each pair can be combined with another pair to give 12 complete determinations of the intensity and direction. The mean intensity and direction are determined and the scatter in intensity values and of the individual directions about the mean serve as indicators of the internal precision of each measurement (Heller 1977). These parameters were computed and used as additional criteria for the acceptability or rejection of each measurement.

3.2 AF DEMAGNETIZATION CHARACTERISTICS

Magnetic cleaning was carried out by progressive alternating field demagnetization using a two-axis tumbler. Measurements of the natural remanent magnetization (NRM) and of the remanences after treatment in peak alternating fields of 60, 100, 150 and 200 Oe were made for every specimen. The NRM varied in intensity from about 5×10^{-7} to 1×10^{-5} G and contained a large low-coercivity component. This soft component was largely removed in low fields of 100 Oe or less, after which the reduction of intensity with increasing magnetic field was more gradual (Fig. 2). By 300–400 Oe the intensity of most samples had fallen below 5×10^{-8} G and could not be adequately distinguished from the instrumental noise. This placed a limit on magnetic cleaning. However, in most samples the soft component was easily remagnetized and anhysteretic components made the use of fields stronger than 200 Oe of dubious value.

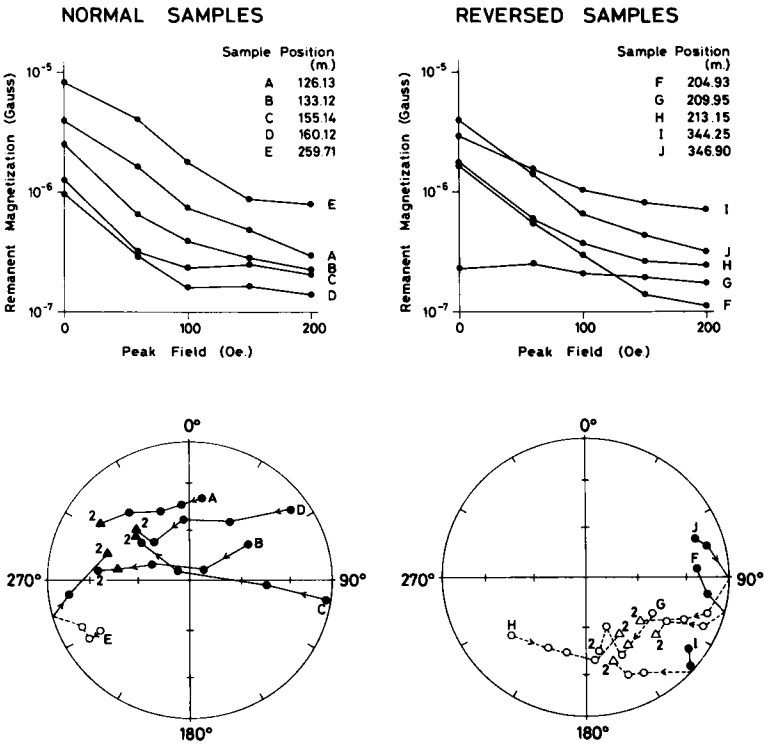


Figure 2. Behaviour during alternating field demagnetization of normal and reversed samples having a large unstable component. Letters A to J identify the NRM direction of the appropriate sample, numeral 2 identifies the direction after 200-Oe demagnetization and the triangles represent the directions which give minimum dispersion. Solid circles are plotted on the lower hemisphere, open circles on the upper hemisphere.

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A major problem in analysing the 493 AF demagnetization curves was how to allow for the presence of the large soft component. The NRM directions, while not random, were generally uninterpretable, and large directional changes took place, usually systematically, during most AF demagnetizations. The directions of normal samples (Fig. 2) tended to move towards the NW quadrant on the lower hemisphere; the directions of reversed samples moved towards the SE quadrant on the upper hemisphere, frequently from initial directions on the lower hemisphere, indicating removal of a soft normal secondary component.

3.3 DETERMINATION OF REPRESENTATIVE DIRECTION

In deciding which direction to accept for each sample use was made of (a) vector diagrams, (b) a stability index and (c) a minimum dispersion criterion.

(a) Vector diagrams were found to be useful for analysing the directions of Scaglia Rossa samples from elsewhere in Umbria (Lowrie & Alvarez 1975). However, in the Bottaccione samples the soft component was too great, and clearly defined straight lines towards the origin could only be drawn for a comparatively small number of samples.

(b) Several palaeomagnetic stability indices have been proposed that test the variations of intensity, direction or both during demagnetization (Tarling & Symons 1967; Briden 1972; Symons & Stupavsky 1974). These indices serve as an aid to selecting the optimum demagnetizing field for a specimen. A more sophisticated index (Giddings & McElhinny 1976) includes the analysis of coercivity spectra in its definition.

For the present study, the variation of remanent direction was of primary interest. A stability index was used which is similar to (but somewhat simpler to compute than) that of Symons & Stupavsky (1974).

The direction cosines of a direction (D_i, I_i) are $l_i = \cos D_i \cos I_i$, $m_i = \sin D_i \cos I_i$ and $n_i = \sin I_i$. The angle between two successive vectors (i and j) on a demagnetization curve is given by:

$$\delta\theta = \cos^{-1}(l_i l_j + m_i m_j + n_i n_j).$$

This angle is divided by the increment in demagnetizing field (δH) to obtain the average angular rate of change during the demagnetization step. The most stable direction is where $(\delta\theta/\delta H)$ is minimum.

The use of such a stability index to interpret demagnetization curves is discussed by Symons & Stupavsky (1974). We used the index to reject a few unusable samples with $(\delta\theta/\delta H)$ greater than 1 deg/Oe, and to select a characteristic direction in more than half the Gubbio samples. However, the stability index of each of the remainder was essentially constant within the limits of experimental error and therefore did not uniquely identify an optimum direction for those samples. For them a minimum dispersion technique was used.

(c) The dispersion of palaeomagnetic directions about their mean is measured by the circular standard deviation of the distribution. Optimum magnetic cleaning is often assumed to be achieved when the dispersion, computed by combining directions at a common hierarchical level of demagnetization, is minimum. In the Gubbio samples minimum dispersion (33.1°) is achieved at 200 Oe. However, this blanket treatment assumes that all samples have the same coercivity spectrum, which in the Gubbio samples is incorrect (Fig. 2). The dispersion can be reduced by substituting directions at different levels of demagnetization for some samples. For example, the direction of Sample B after 150 Oe improves the clustering of the normal samples, while that of sample G after 60 Oe improves the negative group.

The demagnetization path of each sample that did not have a clearly defined minimum stability index was inspected as in Fig. 2, and a representative direction was selected such

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Table 2. (a) Statistical analysis of optimally cleaned remanence directions of all samples from the Gubbio section, and comparison of the means of normally and reversely magnetized samples. Intermediate directions and data from the disturbed zone at 220–230 m have been omitted from the last two groups. (b) IRM acquisition curves were used to classify 51 samples on the basis of whether they contained haematite or not. The mean directions of the two groups are not significantly different.

	Number of samples	Mean direction		α_{95}	R	Angular standard deviation
		Declination	Inclination			
A. All samples	493	305	43	2.6	420.1	31
Gubbio Long Normal Zone	164	290	44	3.1	152.2	22
Above Gubbio Long Normal Zone:						
Normal	114	312	44	5.1	99.6	29
Reversed	167	144	−40	3.7	149.6	26
B. Haematized samples	42	304	47	9.7	35.32	14
Magnetite samples	9	292	45	9.0	8.764	33

that the combined set of (stable and unstable) OPT directions had minimum dispersion. This dispersion (30.7°) was smaller than after any single stage of demagnetization. The improvement over the 200 Oe directions is not very large but is brought about mainly by discrimination against intermediate directions. As a result, although short duration field excursions may be lost from the record, the boundaries of polarity zones are more sharply defined.

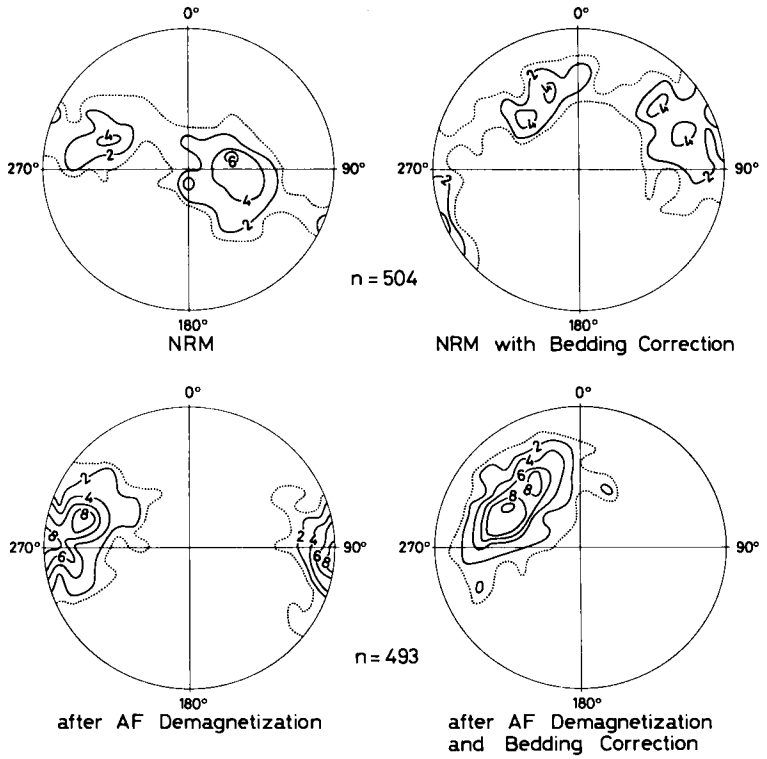


Figure 3. Equal area density maps of the directions of the NRM and of the optimally demagnetized directions, before and after applying tectonic corrections. The figures are all lower hemisphere plots, and the directions of samples with negative polarity have been reversed accordingly. Contour labelling: percentage of data points per 1 per cent of area. Dotted contour line = 1 per cent.

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In previous work on the Gubbio section declinations in the Maastrichtian were found to be about 45° greater than in the pre-Campanian part of the section (Lowrie & Alvarez 1975). The minimum dispersion method makes no allowance for this geological difference between the upper and lower parts of the section, and forces selection of directions closest to the overall section mean in those samples for which it was used.

Normal and reversed OPT directions are present; antipodal directions of reversed samples are used in determining the vector mean directions of groups of samples (Table 2), and in composing equal-area density plots. These demonstrate the improvement in grouping of OPT directions compared to the NRM directions, before and after applying a tectonic correction for the bedding attitude at each sample level (Fig. 3). The directions conform to a Fisherian distribution relative to their mean, but are non-uniformly distributed in azimuth about the mean. This is evident from the oval shape of the cleaned, corrected OPT directions, and reflects predominantly the above-mentioned variation with stratigraphic position in the section.

In spite of a relatively large amount of scatter, the NRM and OPT magnetization intensities appear to show a fairly rhythmic variation with a wavelength of 50–80 m (Fig. 4). The sedimentation rate in the Gubbio section is not constant (Table 1). However, the appropriate period of the fluctuation appears to be about 3–8 Myr. On the basis of presently available data it cannot be established whether this fluctuation is a rock-magnetic property or reflects long-term geomagnetic field behaviour. The variation of stability throughout the section is portrayed by two parameters. The ratio (f_{100}) of the remanent intensity after 100 Oe demagnetization to that of NRM is a measure of the size of the soft fraction; low values indicate instability of intensity. The value of the stability index ($\delta\theta/\delta H$) of the optimally cleaned direction is a measure of directional stability; high values indicate instability. The variations of these two stability parameters in the Gubbio section are not directly correlated with either lithology or stratigraphic position (Fig. 4). The bulk of the section is characterized by f_{100} values in the range 0.1–0.3, although isolated samples have values as high as 0.9. The lower cherty member has more anomalously high stabilities than the chert-free member. The directional stability index ($\delta\theta/\delta H$) for most of the section was in the range 0.2–0.5 deg/Oe, corresponding to low stabilities.

The presence of a large soft component, and the low directional stability in comparison to some igneous rocks, makes analysis of the limestone directions difficult. Although in some cases the quality is marginal, the magnetic properties of the Scaglia Rossa limestone are usually adequate for palaeomagnetic investigation. This conclusion is supported by a positive fold test, and by study of the rock magnetism.

3.4 FOLD TEST

The bedding in the Gubbio section dips homoclinally at 40–60° to the north-east. Only the NE limb of the Gubbio anticline is present, the SW limb being down-faulted beneath the Gubbio plain. The fold axes of the long Umbrian anticlinal ranges are almost horizontal, so a tectonic correction to the measured remanence directions probably can be made unambiguously. However, the classical fold test of the origin and stability of the magnetization (Graham 1949) requires samples from beds with distinctly different attitudes. As part of a programme to support and check our work in the Gubbio section, we have sampled other Umbrian sections. One of these is located at Moria, about 16 km from Gubbio and on the SW limb of a parallel anticline. This section also has been studied palaeomagnetically as part of the derivation of a polar wander path for Umbria (VandenBerg & Wonders 1976). The bedding dips at 30–50° to the south-west and allows a fold test to be made in conjunc-

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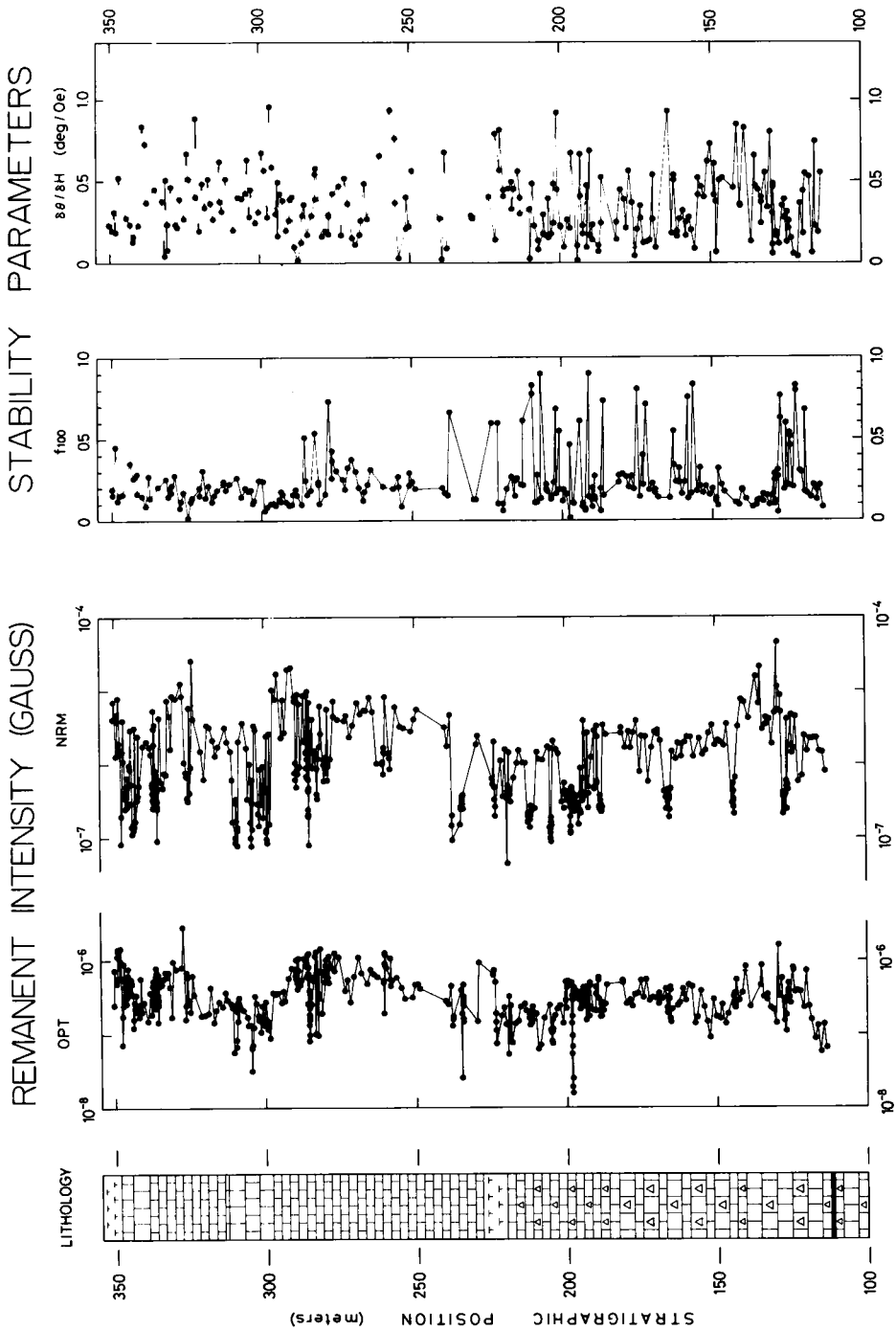


Figure 4. Variations of NRM and optimally cleaned (OPT) remanence intensities, an intensity stability parameter (f_{100}), and a directional stability parameter ($\delta\theta/\delta H$) with stratigraphic position. f_{100} is the ratio of the intensity of remanence after 100-Oe AF demagnetization to that of NRM; ($\delta\theta/\delta H$) is the rate of change of the OPT direction with demagnetizing field. The lithology and measured section are taken from Arthur & Fischer (1977).

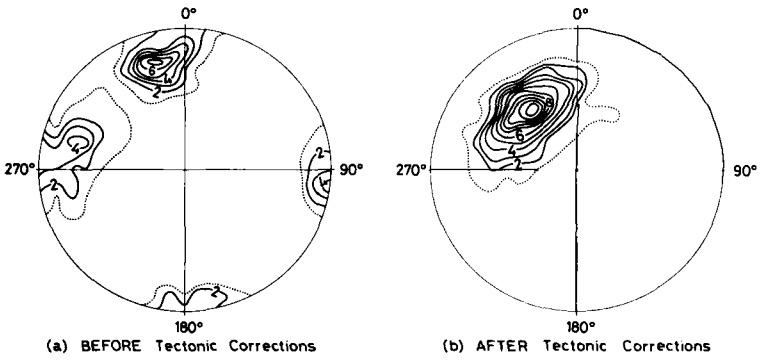


Figure 5. Application of the tectonic fold test. (a) Before applying bedding corrections; directions from the Gubbio section, where the bedding dips to the north-east, lie near 90 and 270°, while those from the Moria section, where the bedding dips southwestwards, lie near 0 and 180°. (b) After applying bedding corrections. Contour labelling: as in Fig. 3.

tion with the Gubbio results. For this purpose we can consider the Gubbio and Moria sections to be located on opposite flanks of a broad, complex syncline.

The stable remanent directions of 250 samples from the Moria section and of an equal number of (alternately selected) samples from the Gubbio section are shown in Fig. 5(a) and (b), before and after applying the appropriate local tectonic corrections, respectively. The dramatic improvement in coherence of the data for the two sections after making these corrections indicates that the remanent magnetization of the Scaglia Rossa predates the major folding of the Umbrian sequence during the late Tertiary, and lends credibility to the palaeomagnetic significance of the tectonically corrected directions from the Gubbio section.

4 Rock magnetism

In an earlier study (Lowrie & Alvarez 1975) we performed a variety of tests of the origin of the remanent magnetization of the Scaglia Rossa. Ferrimagnetic minerals were extracted from three large blocks of the limestone and identified by Curie point analysis as magnetite and haematite. Acquisition of isothermal remanent magnetization (IRM) in selected samples from 37 sites also indicated the presence of either magnetite alone or of magnetite together with haematite. Thermal demagnetization of the natural remanent magnetization (NRM) of samples containing both haematite and magnetite indicated that the direction of the haematite component was close to that of the magnetite. We concluded that the haematization had not drastically altered the primary direction of remanent magnetization of the Scaglia Rossa.

Further tests of Scaglia Rossa samples from Gubbio and elsewhere in Umbria support this conclusion. These tests involve acquisition of isothermal remanent magnetization (IRM) in extremely high fields up to 50 kOe in stratigraphically distributed samples from the Gubbio section, and comparison of the directions of normally and reversely magnetized samples.

4.1 ACQUISITION OF IRM IN GUBBIO SAMPLES

Samples were selected at approximately 5-m intervals through the 250-m thick Upper Cretaceous part of the Gubbio section. An electromagnet was used to give each sample a

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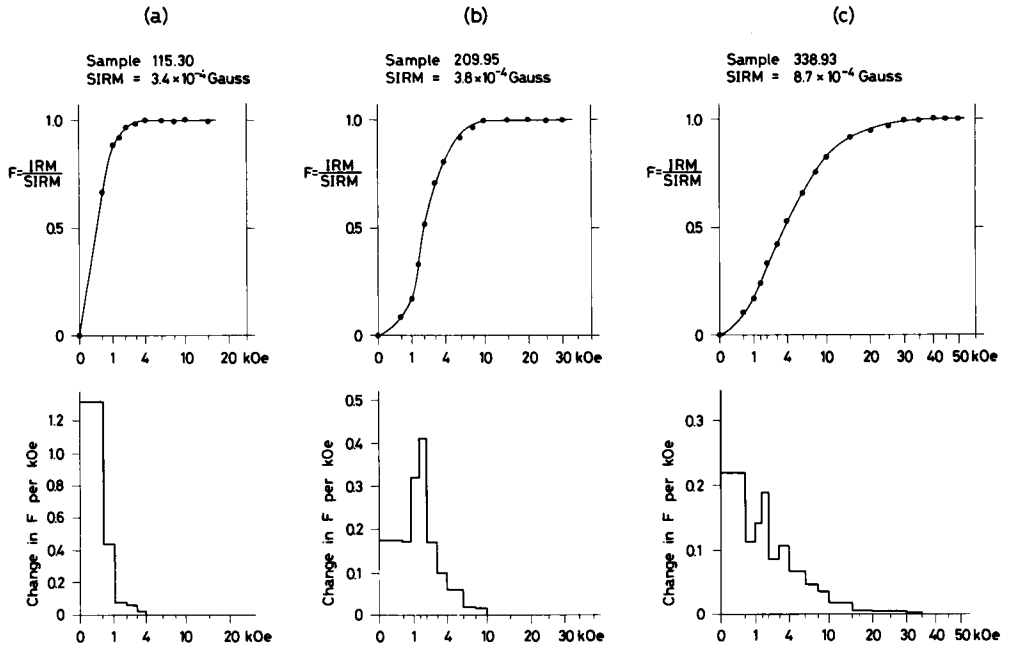


Figure 6. Normalized IRM acquisition curves (above) and coercivity spectra (below) for three types of Scaglia Rossa samples, which achieve effective saturation in (a) fields of 3 kOe or less, (b) fields between 3 and 10 kOe and (c) fields stronger than 10 kOe.

succession of IRMs in progressively increasing fields up to 25 kOe. A superconducting magnet was then used to produce IRM in fields up to 50 kOe. Representative IRM acquisition curves are shown in Fig. 6.

Three types of behaviour were observed. A few samples acquired saturation isothermal remanent magnetization (SIRM) in fields of 3 kOe or less – Fig. 6(a). Most samples reached SIRM between 3 and 10 kOe – Fig. 6(b). The third category of samples required fields greater than 10 kOe to produce SIRM, Fig. 6(c), and a few of these did not saturate in 50 kOe. Their magnetization continued to increase very slowly but discernibly up to the maximum field. Following Dunlop (1972) we identify the magnetic mineral that saturates below 3 kOe as magnetite, we tentatively identify the mineral that saturates in the 3–10 kOe range as specular haematite and the third extremely high coercivity mineral as pigmentary haematite. More positive identification requires thermal analysis. Each coercivity spectrum is dominated by the magnetite fraction; even the samples containing haematite achieved over 60 per cent of SIRM in 3 kOe.

In order to display the variation in the relative amounts of haematite and magnetite in the Gubbio section, the field required to produce SIRM in each of the samples was determined. Similarly, from each IRM curve, the ratio of the IRM acquired in 1 kOe to SIRM was computed. For a sample dominated by magnetite which attains SIRM in fields less than 3 kOe, this ratio is close to unity; for a sample dominated by haematite it is small.

The variations of SIRM and of IRM (1 kOe)/SIRM with stratigraphic position (Fig. 7) illustrate that haematite is an important component of the magnetic mineralogy in most of the Gubbio section. In only two comparatively short sections (between 115 and 120 m, and between 145 and 180 m) and in an isolated sample at 205 m, is haematite absent or insignificant. Systematic investigation of sample colours showed that all the samples in which haematite was missing were white, while all samples containing haematite were pink or red,

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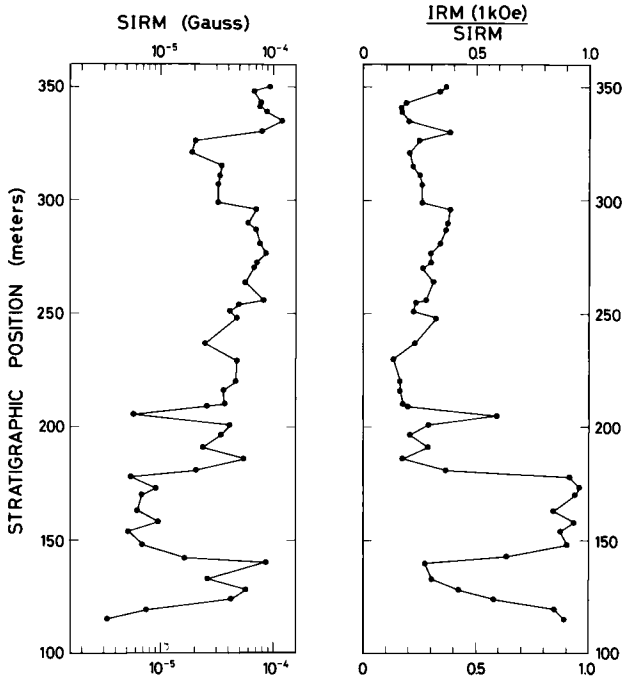


Figure 7. Variation of saturation IRM and of the ratio of IRM produced in 1 kOe to saturation IRM, as functions of stratigraphic position in the Gubbio section.

darkening in hue towards the top of the section (J. E. T. Channell, private communication). However, comparison of Figs 4, 7 and 8 show that there is apparently no correlation between the presence of haematite and either the stability or direction of the natural remanent magnetization. The characteristic remanent magnetization directions of the samples whose IRM curves indicated the presence of little or no haematite were averaged and compared with those of the remainder of the samples whose IRM acquisition was investigated (Table 2). The mean directions are only 8.5° apart and are not significantly different.

A possible interpretation of this result is that although the haematite present may be responsible for the pink colouration it does not contribute appreciably to the NRM, or its contribution may be insignificant in comparison to that of the magnetite. This would be possible if the haematite is of detrital origin, as the process of magnetically aligning the weakly magnetized grains would be very inefficient by either the depositional or post-depositional mechanism. It is more likely, however, that the haematite forms diagenetically from goethite or by oxidation of detrital magnetite and carries a chemical remanent magnetization (CRM). Because of the apparent absence of directional error associated with it, we conclude that this haematization took place very soon after, or contemporaneously with, the post-depositional acquisition of primary remanence by the detrital magnetite, so that both the magnetite and haematite remanent components record the same geomagnetic field direction.

4.2 COMPARISON OF NORMAL AND REVERSED MAGNETIZATION DIRECTIONS

The presence of haematite is potentially serious for proper magnetic cleaning as secondary components of remanence carried by single domain haematite are too hard to be removed by

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alternating field demagnetization. It is possible that the final stable direction may not have been reached in some samples, and a systematic uncleaned component may persist.

A method of testing the effect of inadequate magnetic cleaning is to compare the directions of normal and reversely magnetized samples. In order to allow for the effect of declination variation throughout the Gubbio section (Lowrie & Alvarez 1975) all samples from the Gubbio Long Normal Zone were averaged, and the remaining samples were divided into two groups corresponding to all normal samples and all reversed samples, respectively, from the alternating polarity zones higher in the section. Comparison of these last two groups (Table 2) shows that their mean directions are not antipodal to each other, but differ by 10 degrees. Using the test of Watson (1956), the normal mean is found to be significantly different from the antipode of the reversed mean. This indicates that a systematic stable component, different from the characteristic remanent direction, has not been removed by the magnetic cleaning.

The imperfect isolation of the stable component of magnetization is believed to have only minor effect on the directions of our Upper Cretaceous samples. However, unusual cleaned directions were reported in samples from the Upper Maastrichtian and Palaeocene parts of the Gubbio section (Premoli Silva, Napoleone & Fischer 1974). In this case the true directions were distorted by the presence of a hard secondary magnetization that was successfully removed only by thermal demagnetization (Roggenthen & Napoleone 1977). The direction of the hard secondary magnetization was described as 'a flat vector pointing to the SW'. This presumably is a late overprinted remanent component that is not important in the Upper Cretaceous part of the Gubbio section, but which acquires increasing importance near the Cretaceous–Tertiary boundary and in the Palaeocene.

4.3 SUMMARY: THE ORIGIN OF NRM IN THE UMBRIAN SCAGLIA ROSSA LIMESTONE

The rock magnetic studies have established that magnetite is probably present throughout the Gubbio section, but in most of the section haematite is also present. The demagnetization stability studies showed that identification of a stable direction is complicated by the presence of a very large soft component. The characteristic remanence (OPT) of the Scaglia Rossa is identified with an intermediate stability component, and its direction is modified slightly by the presence of a component that was not completely removed by AF cleaning to 200 Oe.

These characteristics are consistent with the magnetic properties of the Scaglia Rossa limestone elsewhere in Umbria (Lowrie & Alvarez 1975). The NRM is identified with the magnetite fraction and is assumed to be acquired by a mechanism similar to that associated with the magnetization of modern deep sea sediments.

Primary remanent magnetization in sediments is often thought of as depositional remanent magnetization (DRM) acquired by alignment of detrital magnetite particles with the magnetic field during slow settling (King 1955; Griffiths *et al.* 1960). This model may be true in coarse or medium-grained sediments but is not applicable for fine-grained pelagic deep sea sediments. Irving & Major (1964) described a method by which fine-grained sediments could acquire a post-depositional remanence (PDRM) in the soft sediment state by alignment of interstitial magnetic grains with the ambient field. This has been shown to be a viable mechanism for the remanent magnetization of magnetite-bearing pelagic deep-sea sediments, in which the remanence becomes progressively locked into the sediment some 10 cm or so below the sediment–water interface during partial compaction (Løvlie 1974, 1976). This type of remanence has no inclination error (Kent 1973), in agreement with

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what is known about the inclinations of modern deep sea sediments (Opdyke & Henry 1969). It is reasonable to assume that the Scaglia Rossa remanence is of this type and that it can be interpreted as a direct record of the palaeomagnetic field directions during deposition.

The implications of a possible time-lag between sediment deposition and acquisition of remanence must be taken into account in interpreting magnetic stratigraphy. The deposition rate of the Scaglia Rossa at Gubbio averages about 1 cm per 1000 yr (Arthur & Fischer 1977). A typical time lag of 10 cm such as that reported by Løvlie (1974) would correspond to about 10 000 yr. It should therefore be possible to observe magnetic stratigraphy in the Scaglia Rossa with as good resolution as in most Pleistocene and Pliocene deep sea sediments. Short events lasting a few tens of thousands of years (equivalent to the well-known Jaramillo event) would survive, but events of shorter duration than the time-lag would be obliterated, if the remanence acquisition front is progressive as shown by Løvlie (1976).

Bioturbation would also affect this time-lag. The average mixing depth in unconsolidated deep-sea sediments has been reported as 10 cm (Guinasso & Schink 1975) with occasional penetration by burrowing organisms to several times this depth. Arthur & Fischer (1977) estimate that burrowing after black shale deposition extended from 3 to 15 cm below the surface and was usually less than 5 cm in the compacted limestones (equivalent to about 10 cm of uncompacted sediment). The bioturbation effect would produce an effective time-lag in the acquisition of remanent magnetization approximately equivalent to that of the post-depositional acquisition mechanism.

5 Magnetic stratigraphy

The inclinations and declinations of the optimum set of magnetizations were plotted against stratigraphic position (Fig. 8). Both parameters show considerable scatter; whether this is due to inhomogeneity of the magnetization because of the low magnetic mineral concentration (probably around 0.01 per cent, Lowrie & Alvarez 1975), or to disturbance because of bioturbation has not been established, although all of these probably contribute to some degree. Polarity fluctuations are shown by almost synchronous, sharp transitions of both angles at each reversal boundary. They are not related to lithology variations, but there are three zones in which tectonic effects have to be taken into account.

At approximately 198 m a steep fault offsets the moderately dipping beds near the end of a road-cut. Samples from above the fault show a reversal boundary just above it. The section was continued in the wall of the gorge back from the road where the fault was again located. Samples from below the fault here showed a well-defined reversal boundary that is apparently equivalent to the boundary immediately above the fault (Fig. 8). We could not recognize any characteristic chert bed offset by the same amount as the reversal boundary and although it is conceivable that one or more pairs of very thin polarity zones could have been cut out by the fault, the same fossil zone is found on either side of the fault, and it is clear that no major stratigraphic interval is missing.

A short disturbed section marked by a very tight fold, a thick, white, possibly turbiditic bed and a sequence of thin shaly beds was sampled at 220–228 m near the top of a negative polarity zone. This short section is tectonically unreliable and these data must be excluded from further interpretation. Immediately above is a short section with fairly reliable bedding and negative polarity. Above that is a long covered interval from 230 to 248 m in which isolated outcrops have similar bedding attitudes to the rest of the Gubbio section, but where the continuity of the section cannot be followed. The sampling interval is very wide here;

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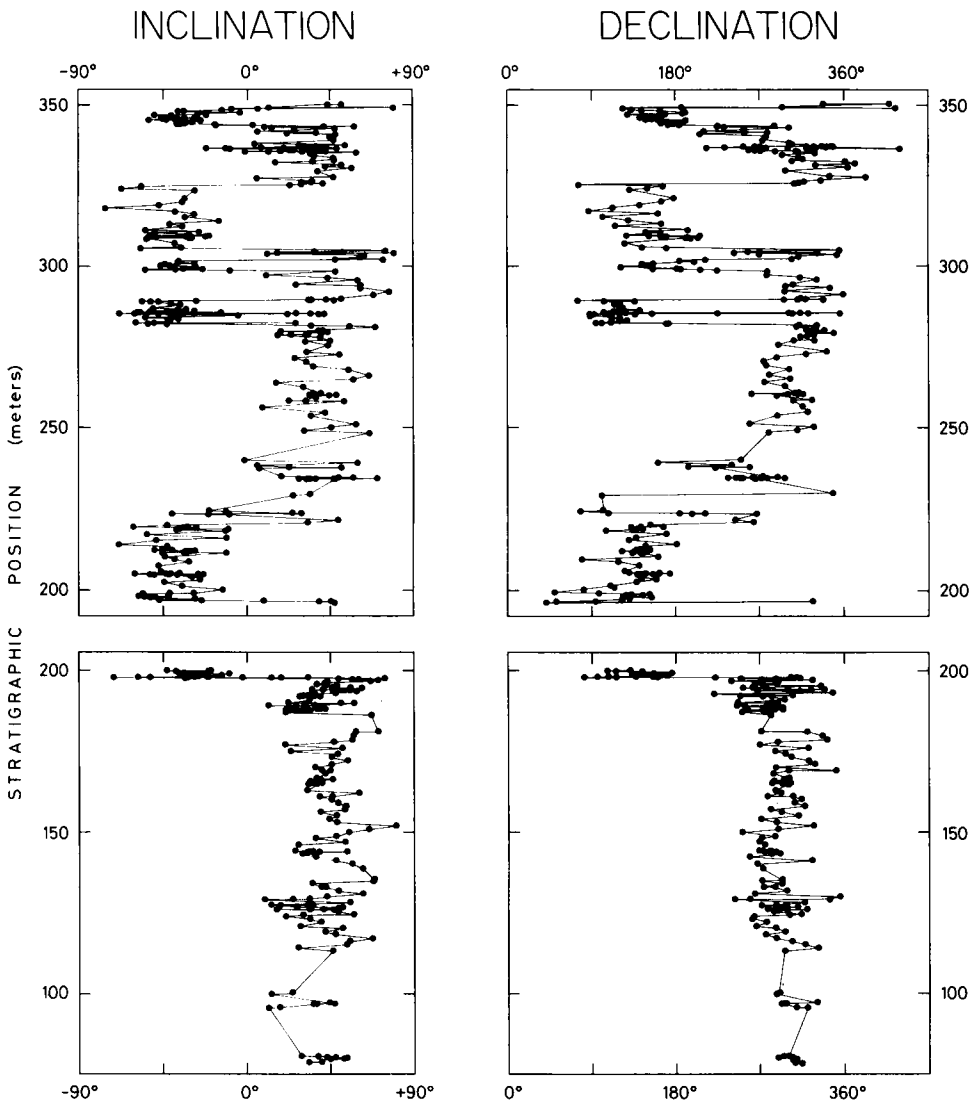


Figure 8. Variations of declination and inclination of optimally cleaned samples with stratigraphic position. A fault at about 198 m offsets the two halves of the section but the same magnetic polarity transition is seen above and below it.

the polarities of the samples are probably safely identified, although the data from this interval must be regarded with caution for more detailed analysis.

The inclination of the mean direction of the Scaglia Rossa in the Gubbio section (Table 2) is too low and indicates that deposition occurred at a much lower latitude than the present site latitude of 43°N. The latitude of deposition was computed from the inclination of the mean direction to be 25°N. The declinations expected at Gubbio on the basis of comparison with the Van der Voo & French (1974) estimated Late Cretaceous poles for the European and African plates should be 345–355°, whereas the mean observed declination is 305° (Table 2). This discrepancy of 40–50° was earlier interpreted as evidence for counterclockwise rotation of the Italian peninsula (Lowrie & Alvarez 1974).

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The virtual geomagnetic pole (VGP) latitude was computed for each sample. The 25° depositional latitude based on the observed mean inclination was assumed in the calculations, and the individual declinations were corrected by rotating each back clockwise so that the mean corresponded to 355°. The variation of VGP latitude with stratigraphic position is shown in Fig. 9.

The polarity sequence was determined from the VGP latitudes. Except for the short disturbed section and the longer covered interval where there are large gaps between data groups, the Gubbio section shows a well-defined polarity sequence for the Late Cretaceous (Lowrie & Alvarez 1977; Lowrie, Alvarez & Premoli Silva 1976). Although there is appreciable scatter of VGP latitude throughout the section, distinct zones of normal and reversed polarity are easily identified.

In the course of establishing a type section at Gubbio, Alvarez *et al.* (1977) have identified each of the polarity zones by letter, the subzones by number and the corresponding polarity by its sign (Fig. 9). A long zone of continuous normal polarity at the base of the section is identified as the Gubbio Long Normal zone.

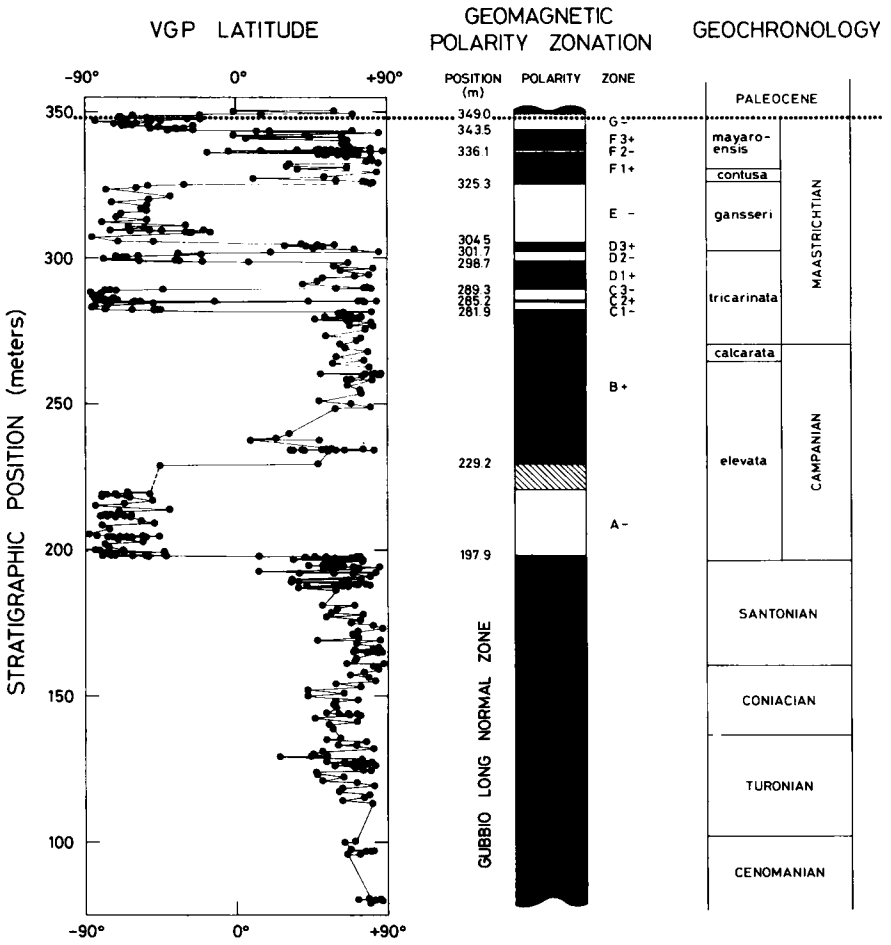


Figure 9. Late Cretaceous geomagnetic polarity sequence derived from VGP latitudes. The lettered zones and numbered subzones follow the classification system of Alvarez *et al.* (1977). The locations of the geochronological stage boundaries are from Premoli Silva (1977) and Premoli Silva *et al.* (1977).

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The Late Cretaceous geomagnetic polarity sequence from the Gubbio section has been shown to agree very closely with that obtained from marine magnetic anomaly profiles in the North Pacific, North Indian and South Atlantic oceans (Lowrie & Alvarez 1977). The main features of the correlation (Fig. 9) are the following:

(a) The Gubbio Long Normal zone is identified with the Cretaceous magnetic quiet zone, confirming the interpretation of this quiet zone as representing a period of constant geomagnetic polarity.

(b) The Gubbio Long Normal Zone ends with the reversed zone Gubbio A−. The equivalent block of reversely magnetized oceanic crust is the source of magnetic anomalies 33 and 34 (Larson & Pitman 1972). The existence of these anomalies had not been accepted universally but is confirmed by the Gubbio section, which correlates the end of the Cretaceous quiet zone at anomaly 34, close to the Campanian–Santonian boundary (Fig. 9).

(c) The Cretaceous–Tertiary boundary, located at the base of the *Globigerina eugubina* zone at 347.6 m (Premoli Silva 1977) falls within reversed zone Gubbio G− equivalent to the interval between anomalies 29 and 30, and precedes the nearest reversal.

The additional detailed sampling included in the present paper refines the above picture by giving the location of each reversal boundary to within 0.2 m, except for that associated with anomaly 33. The structural complications discussed above first showed up as anomalous magnetic directions in the detailed sampling data. As a result, it is felt that the exact location of this boundary could not be determined, and it is shown as a hatchured zone in Figs 9 and 10. The polarities are safely identified in the underlying reversed zone and in the overlying normal zone.

6 Discussion and conclusions

The rock magnetic evidence presented by Lowrie & Alvarez (1975) and in this study confirm that the magnetic properties of the Scaglia Rossa limestone in Umbria are satisfactory for a palaeomagnetic study. In spite of the weakness of the magnetization it can be readily and accurately measured with modern magnetometers. The stability is low but adequate to preserve a good record of the geomagnetic reversals in the Upper Cretaceous.

The application of palaeomagnetism to rocks from Umbria is clouded by the unresolved question of autochthony of the Umbrian formations. There are few published works on the subject and geological opinion is divided. The region was described as autochthonous or parautochthonous by Bortolotti *et al.* (1970). Assuming autochthony, westerly declinations encountered in previous palaeomagnetic studies in Umbria have been interpreted as evidence for tectonic rotation of the Italian peninsula as an independent microplate (Lowrie & Alvarez 1975) or as part of the African plate (VandenBerg *et al.* 1977).

In other studies (Channell & Tarling 1975; Channell & Horváth 1976) the Umbrian Apennines were presumed to be allochthonous. Triassic evaporites have been encountered in some deep wells and could provide a décollement surface decoupling the limestone sequence from its basement. Such a décollement, if it exists, could have affected declinations in Umbria and could account for the observed westerly declinations in the Gubbio section. However, the magnetic stratigraphy of the Gubbio section is valid regardless of the plate tectonic history of Umbria.

There is almost complete agreement between the polarity sequence recorded in the 250-m thick vertical section of the Scaglia Rossa limestone at Gubbio and the sequences recorded in the oceanic crust over horizontal distances hundreds of kilometres in length in different ocean basins (Fig. 10). The lengths of the normal and reversed zones in the Gubbio section do not represent directly the durations of the corresponding geomagnetic polarity epochs

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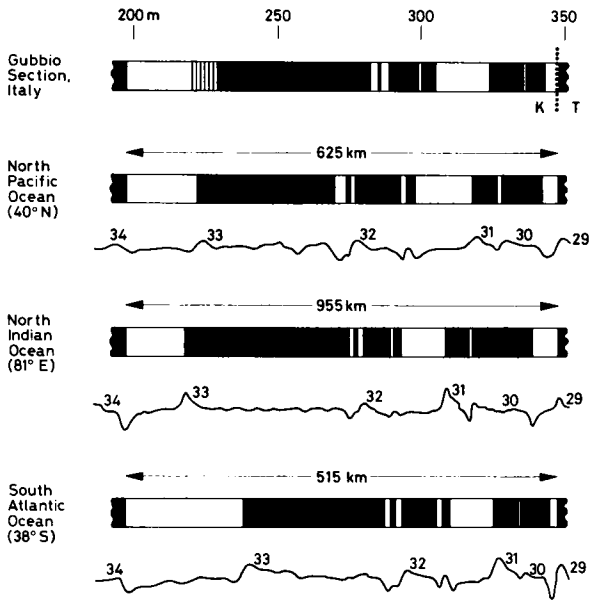


Figure 10. Comparison of the Gubbio polarity sequence from the 250-m thick Scaglia Rossa limestone with Late Cretaceous geomagnetic polarity sequences derived from oceanic magnetic anomaly profiles extending over hundreds of kilometres of horizontal distance. Profiles are identified in Lowrie & Alvarez (1977, Fig. 2). Reversal boundaries on all three profiles were located by R. L. Larson (1975, private communication).

because the sedimentation rate is not constant (Table 1). The evidence from the oceanic anomalies indicates that along at least two of the profiles non-linear spreading was also taking place.

The interpretation of the marine magnetic anomalies (R. L. Larson 1975, private communication) predicted a short normal zone in the middle of reversed zone C−. This sub-zone C2+, was not observed by Lowrie & Alvarez (1977). Detailed sampling of the C− interval was undertaken to investigate its existence; it was found where predicted as a very thin, but unambiguous, interval of normally magnetized samples (Figs 8, 9, 10).

The most doubtful polarity zone in the sequence is the short reversed subzone F2− which, however, was reported also by Roggenthen & Napoleone (1977). It is incomplete and suggests a magnetic 'excursion'. This feature is seen on all of the oceanic profiles as a small undulation (Fig. 10). Such 'tiny wiggles' have been variously interpreted as short events and as fluctuations in intensity of the geomagnetic dipole field. Because of the probable filtering effects of time lags caused by bioturbation and the remnanence acquisition mechanism it is likely that a feature as short-lived as an excursion would be obliterated. The Gubbio record does not show any special intensity variation associated with F2−, and it seems safest to interpret this as a short event lasting one or two tens of thousands of years, perhaps, which is partly obscured by the filtering effects.

In the Gubbio section foraminiferal extinctions locate the Cretaceous–Tertiary boundary in the Gubbio G− zone between anomalies 29 and 30. Butler *et al.* (1976) have studied the magnetic stratigraphy of Upper Cretaceous–Lower Tertiary dinosaur-bearing terrestrial sediments from Arizona and report a good agreement with the oceanic anomalies sequence. They locate the Tertiary–Cretaceous boundary (based on dinosaur extinctions) in the *normal* zone corresponding to anomaly 29, i.e. distinctly younger than in the Gubbio

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section. If this observation is substantiated by full evaluation of their data, the difference between their section and the Gubbio section could indicate either an unrecognized hiatus in one section or the other, or that extinction of forams and dinosaurs are not synchronous. If the latter possibility is borne out by future work, it will be an extremely important clue for the eventual resolution of the question why these extinctions occurred.

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